7.1 A brief introduction to GraphQL

GraphQL (https://graphql.org/) is an API standard that provides a more efficient, powerful, and flexible alternative to REST. Facebook initially developed it, but it's now maintained by a large open source community.

Unlike REST APIs, GraphQL allows the request to describe the desired structure of the response, which is why a single API endpoint can be used for multiple purposes. To explain the difference, let's consider the scenario of fetching data. In the context of our movie application, let's say the consumer wants to create a landing page for each movie. The landing page will have data about the movie, the different actors from the movie, and other movies they have starred in. With a REST API, because an endpoint returns a fixed set of data structures, consumers would need to make different `GET` requests to multiple endpoints to retrieve all the data that they need, as illustrated in figure 7.1.